When Explanations Outlive Their Data: Faithfulness Decoupling in Graph-Attention MARL Fleet Dispatch under Telemetry Degradation

A Master's Dissertation

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By

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DECLARATION

I declare that this dissertation is my own original work carried out under the supervision of Farhan S. Ujager and that all sources and references have been acknowledged appropriately. This dissertation has not been submitted previously for academic credit at De Montfort University, Dubai, or any other institution.

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Date: ______________________________

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LIST OF ABBREVIATIONS

Abbreviation	Full form
AoI	Age of Information
CTDE	Centralised Training with Decentralised Execution
DEF	Decision-level Explanation Faithfulness
GAT	Graph Attention Network
MAPPO	Multi-Agent Proximal Policy Optimisation
MARL	Multi-Agent Reinforcement Learning
SUMO	Simulation of Urban MObility
WAMSN	Weighted Attention Mass on Stale Nodes
XAI	Explainable Artificial Intelligence

ABSTRACT

Graph-attention multi-agent reinforcement learning (GAT-MARL) can expose attention weights as explanations of fleet-dispatch decisions. The map remains available when vehicle telemetry becomes stale, but does not show whether its evidence is current or decision-relevant. This dissertation asks: **can raw graph-attention weights be trusted as explanations when fleet telemetry is degraded?**

The experiment simulates 20 taxis in SUMO. Tunnel entry triggers signal loss, and the observation layer freezes the affected vehicle's last-known position and speed for 10, 20, 30, or 60 seconds. SUMO retains the true state, allowing paired clean and degraded observations. Clean-trained and degradation-trained GAT policies use three independent training seeds. Decision-level explanation faithfulness (DEF) compares attended nodes with type-matched random nodes, while WAMSN measures stale-data exposure. A construct-validity audit controls for request nodes that also represent actions.

The results do not validate raw attention as a dependable explanation. Under clean telemetry, DEF varies in sign across checkpoints and is lower than a leave-one-out perturbation control in all six policies. Longer outages increase stale-data exposure, but the paired change in attention toward stale nodes is positive for two training seeds and negative for one under both training regimes. DEF does not consistently decline with outage duration, and the result also depends on the layer, head, and query row used for explanation.

The conclusion is not that stale telemetry always reduces faithfulness. It is that raw attention provides no stable guarantee of either data freshness or decision relevance across independently trained policies. An attention map can therefore outlive the data supporting it while still appearing complete. Attention weights should be treated as model internals unless each released checkpoint passes separate freshness and faithfulness checks.

Keywords: explainable reinforcement learning; graph attention; multi-agent reinforcement learning; fleet dispatch; telemetry degradation; Age of Information; faithfulness.

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Chapter 1: Introduction

1.1 Context

Fleet dispatch is relational. A taxi's useful action depends on nearby vehicles, open passenger requests, road conditions, and competition for the same demand. Graph neural networks represent these relationships directly, and graph attention networks (GATs) add learned weights over neighbouring nodes (Scarselli et al., 2009), (Velickovic et al., 2018). Because the weights are easy to display, they are often read as an explanation of which vehicles or requests influenced a decision.

That reading is convenient, but it is not automatically valid. Prior work has shown that an attention weight can be weakly related to the evidence that actually changes a model's output (Jain and Wallace, 2019; Wiegreffe and Pinter, 2019; Serrano and Smith, 2019; Liu et al., 2022). The issue becomes more serious in an operational system because the input itself may no longer describe the current world.

1.2 Problem statement

Vehicle telemetry can be interrupted by tunnels and other signal-loss areas. During an interruption, a dispatch platform may keep the last received position and speed. The reading remains available, but its Age of Information (AoI) grows (Kaul, Yates and Gruteser, 2012). An attention map can therefore continue to look complete even though some of the nodes represent old information. The interface reveals where the model placed attention, but it does not reveal whether that information is fresh or whether removing it would change the decision.

This creates an assurance problem. An operator may read a complete attention map as a trustworthy reason even when its evidence is stale, and stable dispatch performance cannot show whether the explanation is valid. The dissertation is therefore not trying to improve the number of pickups. It is trying to determine whether raw graph-attention weights provide dependable evidence about a decision under clean and degraded telemetry.

The study uses **faithfulness decoupling** for the risk that an explanation remains available and visually plausible after the freshness of its supporting data has expired, without a dependable link between the displayed weights and the evidence that changes the decision. The original hypothesis expected faithfulness to decline while performance remained stable. The final experiment tests this possible pattern, but the broader problem is whether attention offers a reliable assurance at all.

1.3 Research questions

The central research question is:

Can raw graph-attention weights be trusted as explanations of fleet-dispatch decisions when vehicle telemetry becomes stale?

Four operational questions provide the evidence needed to answer it:

1. **RQ1:** Is graph attention a faithful explanation under clean telemetry?
2. **RQ2:** When tunnel-triggered outages occur, does attention move toward stale vehicle nodes?

3. **RQ3:** As outage duration increases, does the relationship between attention, stale-data exposure, and decision relevance remain dependable?
4. **RQ4:** Does training the policy under telemetry degradation make the explanation response more reliable?

The construct-validity audit supports RQ1-RQ3. It is not a separate primary problem. Its purpose is to check that the faithfulness metric measures hidden information rather than the accidental deletion of an available action.

1.4 Objectives

The measurable objectives are to:

- build a reproducible SUMO fleet-dispatch benchmark with train, validation, and held-out test demand;
- train clean and degradation-aware policies across three independent seeds;
- trigger signal loss from tunnel entry while applying the resulting freeze at the observation boundary;
- compare clean telemetry with fixed observation-layer outages of 10, 20, 30, and 60 seconds;
- measure explanation faithfulness with action-aware, type-matched counterfactual controls;
- check whether the faithfulness evaluator responds to an LOO perturbation ranking and quantify the resolution lost through top-k overlap;
- measure how much attention is assigned to stale vehicle information;
- test whether conclusions change across attention layers, heads, rollout, and the query row used as the explanation;
- treat the trained policy, rather than each individual decision, as the unit for judging whether a finding is consistent.

1.5 Main findings

The completed experiment gives one consistent overall answer: **raw attention weights are not validated as dependable explanations by this study**. This is a negative assurance result, supported by three connected findings.

First, clean type-matched DEF varies from -0.0084 to +0.0187 across the audited checkpoints. An LOO perturbation control is larger and positive in every checkpoint, while taxi-only rank agreement with LOO changes sign. The evidence therefore does not support one reproducible account of decision relevance. The supporting audit also shows why a uniform random baseline is misleading when request nodes are actions.

Second, longer outages consistently increase WAMSN. This result appears in all three GAT policies and all three GAT-Outage policies. It shows that more of the displayed attention is attached to stale information when stale exposure lasts longer. It does **not** show that AoI causes lower faithfulness.

Third, the paired change in stale-node attention is not consistent across training seeds. It is positive for seeds 42 and 43 but negative for seed 44 in both GAT and GAT-Outage. H1 and

H2 are unsupported in every trained policy. The association between WAMSN and DEF is supported for two GAT-Outage seeds, but not the third and not any GAT seed. This disagreement is not an absence of a conclusion: it shows that raw attention has no stable, model-independent response to stale telemetry.

The result is also sensitive to how attention is turned into one explanation. Individual heads can reverse the stale-attention direction within a checkpoint, and changing from the self row to the selected request row improves one GAT checkpoint but worsens others. These checks make the lack of a default explanation guarantee more precise.

Finally, degradation-aware training does not remove training-seed variation. It also produces one weak policy replicate, which limits any mitigation claim. Taken together, the results do not prove that attention is always unfaithful. They show that attention cannot be trusted by default; freshness and faithfulness must be checked separately for every released policy.

1.6 Contributions

The dissertation contributes:

- a paired clean/degraded benchmark in which SUMO ground truth is separated from the observation received by the policy;
- an explicit distinction between a tunnel trigger and an observation-layer outage;
- an action-aware faithfulness protocol for graphs where request nodes also define available actions, supported by LOO and overlap diagnostics;
- a sensitivity audit showing how layer, head, rollout, and query-row choices affect the reported attention explanation;
- evidence across independently trained policies that stale-data exposure is consistent, while attention reallocation and faithfulness effects are not;
- a reproducible experiment runner, validation-selected checkpoints, preflight checks, machine-readable results, and training-seed synthesis.

1.7 Dissertation structure

Chapter 2 reviews MARL dispatch, graph attention, explanation faithfulness, and AoI. Chapter 3 describes the simulator, models, degradation layer, metrics, and statistical design. Chapter 4 reports the results. Chapter 5 discusses what can and cannot be concluded. Chapter 6 closes with the practical implications for trustworthy fleet-dispatch explanations.

Chapter 2: Literature Review

This chapter develops the research gap in five steps. It first explains why fleet dispatch is a relational reinforcement-learning problem. It then asks what an explanation should provide in sequential decision making, why raw attention is disputed as an explanation, how graph predictions are commonly explained, and why faithfulness tests can themselves be misleading. The final sections connect these issues to stale telemetry and position the experiment.

2.1 Reinforcement learning for fleet dispatch

Fleet dispatch links vehicles, passenger requests, and a road network over time. A decision for one taxi changes the demand available to others, so a collection of independent single-agent policies can create competition or duplicated assignments. Lin et al. (2018) model city-scale fleet management as a cooperative multi-agent reinforcement-learning (MARL) problem. Qin, Zhu and Ye (2022) show that reinforcement learning is used for matching, repositioning, pricing, and route decisions, often under changing supply and demand (Qin, Zhu and Ye, 2022).

MARL offers several ways to represent coordination. MADDPG learns decentralised actors with centralised critics (Lowe et al., 2017). QMIX factorises a team value into agent values while preserving a monotonic relation to the joint action (Rashid et al., 2020). MAPPO uses the more familiar PPO objective with centralised training and decentralised execution; its empirical stability makes it a practical baseline for cooperative tasks (Yu et al., 2022). This dissertation adopts MAPPO because the research focus is the explanation attached to each taxi's local decision, not a new MARL algorithm.

Graphs provide a natural representation of the local dispatch problem. Nodes can represent the acting taxi, nearby taxis, and open requests; edges permit their information to be combined (Scarselli et al., 2009). A graph attention network (GAT) assigns learned weights while aggregating neighbouring nodes (Velickovic et al., 2018). Actor-Attention-Critic similarly uses attention for communication in MARL (Iqbal and Sha, 2019). These weights are easy to visualise, which makes them tempting to present as reasons for an action. Ease of display, however, is not evidence that the weights identify the information that caused the decision.

Recent dispatch methods continue to use relational structure. BMG-Q represents ride-pooling dispatch through a local bipartite matching graph (Hu, Feng and Li, 2025); CoopRide studies cooperation across city grids (Wang et al., 2025); and DualG-MARL combines vehicle-state and task graphs (Sha et al., 2026). Their primary outcomes are dispatch or scheduling quality. They do not test whether graph weights remain valid explanations when vehicle telemetry becomes stale. These studies are therefore useful task benchmarks, but not direct faithfulness benchmarks.

2.2 Explainability in reinforcement learning

Explainable reinforcement learning (XRL) covers more than a visual saliency map. An explanation may describe which state features mattered, summarise a policy, contrast the selected action with an alternative, or show the sequence of events that led to an outcome. Reviews by Heuillet, Couthouis and Diaz-Rodriguez (2021), Puiutta and Veith (2020), and Milani et al. (2024) all stress that the method must match the user's question. A model developer debugging a policy and an operator checking one live action do not need the same output. The systematic taxonomy by Bekkemoen (2024) reaches the same conclusion across a larger body of recent XRL studies.

This dissertation concerns a local, post-decision operator question: *which visible vehicles or requests supported this dispatch action?* Earlier XRL work demonstrates several possible answers. Greydanus et al. (2018) perturb image regions to visualise what changes an Atari policy. Madumal et al. (2020) use a causal model to generate contrastive explanations. Mott et al. (2019) introduce an attention bottleneck intended to expose what an RL agent uses. These approaches differ in mechanism, but they share an important lesson: a useful picture and a faithful account of model dependence are separate properties.

That distinction is especially important in an operational dashboard. A plausible explanation may be easy for a person to read but weakly connected to the policy computation. A faithful explanation should change when information that matters to the action is removed or retained. The experiment therefore does not ask whether users like the attention map. It asks whether its ranking passes a controlled decision-level faithfulness test.

2.3 Is attention an explanation?

The modern debate began with Jain and Wallace (2019), who found that attention weights could be weakly correlated with other importance measures and that very different attention distributions could produce similar predictions. Wiegreffe and Pinter (2019) argued that this does not justify rejecting all attention explanations and proposed more careful tests. Serrano and Smith (2019) likewise showed that removing high-attention items can affect outputs, but that the relationship varies by model and task. The useful conclusion is not a universal yes or no. It is that attention needs an explicit claim and an appropriate test.

Attention became widely visible through Transformer models (Vaswani et al., 2017), but visibility also encouraged a broader interpretability claim than the mechanism alone can support. Lipton (2018) warns that the word *interpretability* often combines several different goals that should be evaluated separately.

Jacovi and Goldberg (2020) separate *plausibility*, which concerns whether an explanation looks reasonable to a person, from *faithfulness*, which concerns whether it reflects the model's actual reasoning process. This dissertation follows that distinction. The attention weights are not rejected because they are internal values, and they are not accepted because they are coupled to the actor. They are treated as a candidate explanation whose decision relevance must be measured.

Layer and head aggregation add another difficulty. A two-layer, multi-head network does not produce one unique attention map. Averaging heads, selecting a layer, taking a maximum, or composing attention across layers can produce different rankings. Abnar and Zuidema (2020) propose attention rollout and flow to account for information mixing across layers, and report stronger correlations with ablation and gradient importance than raw attention in Transformers. Although their model is not a GAT-MARL dispatcher, the methodological point transfers: an attention result is incomplete unless the aggregation rule is defined and its sensitivity is checked.

2.4 Explaining graph neural networks

GNN explanations may identify important features, nodes, edges, or subgraphs. GNNExplainer learns a compact subgraph and feature mask that preserves a prediction (Ying et al., 2019). PGExplainer parameterises explanation generation so that it can be shared across examples (Luo et al., 2020). SubgraphX searches for important subgraphs using Shapley-value approximations (Yuan et al., 2021). These methods show that graph explanation is a distinct problem: removing one node can alter both its own features and the messages available to other nodes.

The survey by Yuan et al. (2023) organises GNN explainers by target, mechanism, and scope. GraphFramEx goes further by comparing explanation methods under different user needs and combining necessity and sufficiency views (Amara et al., 2022). Its results show that no single method dominates every evaluation dimension. This supports the use of more than one diagnostic in the present study: DEF tests counterfactual relevance, taxi-only rank correlation compares attention with leave-one-out effects, and stale-attention measures describe freshness exposure.

Most GNN explainability benchmarks study node or graph classification. Fleet dispatch differs because some graph nodes also define actions. A passenger request is both information and a selectable action, while a nearby taxi is context only. Deleting these node types is therefore not the same intervention. This action-linked graph structure motivates the construct- validity audit in Section 3.7.

2.5 Faithfulness evaluation and its pitfalls

Comprehensiveness and sufficiency are common perturbation measures. In the ERASER benchmark, comprehensiveness asks how much a prediction weakens when the explanation is removed, while sufficiency asks how well the selected evidence alone preserves it (DeYoung et al., 2020). Liu et al. (2022) similarly test attention by measuring faithfulness violations. These measures are useful because they connect an explanation to model behaviour instead of visual appeal.

Model-agnostic methods such as LIME also use local perturbations to estimate which inputs support a prediction (Ribeiro, Singh and Guestrin, 2016). Their usefulness depends on whether the perturbed samples remain meaningful for the model and task.

Perturbation is not automatically valid. Removing input features may create samples unlike the data seen during training. ROAR addresses this issue by removing features and retraining the model, showing why a simple deletion test can confound attribution quality with distribution shift (Hooker et al., 2019). Adebayo et al. (2018) provide a different sanity check: an explanation should respond when model parameters or training labels are randomised. Alvarez-Melis and Jaakkola (2018) show that explanation methods can also be locally unstable. Together, these studies make the evaluator part of the object being audited.

The present action-deletion problem is related but more specific. If a selected request node is masked, its action logit becomes unavailable. A large margin change can then occur even if the request features carried no useful evidence. A random baseline that removes more request nodes than the attention top-k is not a fair comparison. Type matching, selected-action protection, and clamp logging are introduced to control this mechanism. A leave-one-out perturbation ranking acts as a positive control for comprehensiveness, while Gradient x Input provides a comparator that does not use attention coefficients.

The small graph creates a second limit. When k = 3 and only a few valid nodes exist, a random size-three set must overlap substantially with the explanation set. This compresses the possible difference between them. Reporting the exact expected overlap and a taxi-only rank correlation makes that limitation visible instead of treating every near-zero DEF as decisive evidence.

2.6 Age of Information and telemetry degradation

Age of Information (AoI) measures the time since the newest received update was generated (Kaul, Yates and Gruteser, 2012). It is widely used to study communication freshness and the effect of delayed state on estimation or control. The explanation question is different. A dispatch action and its attention map may remain available even when a vehicle reading has stopped updating. The operator can see a strong weight without knowing that its source is old.

This dissertation uses AoI only as a node-level freshness descriptor and as the weight inside WAMSN. The manipulated condition is the declared duration of an observation-layer outage triggered by tunnel entry. The design does not manipulate AoI independently, so it does not claim that a larger AoI value causes lower faithfulness. Instead, it asks whether stale exposure and attention behaviour remain separately trustworthy under controlled outages.

2.7 Research gap and positioning

The reviewed work establishes four points: relational RL is appropriate for fleet dispatch; attention is easy to expose but disputed as an explanation; GNN faithfulness requires graph-aware perturbations; and stale state matters to operational control. What is not joined is the assurance question at their intersection.

Within the graph- and RL-explainability studies reviewed in Sections 2.2-2.5, no evaluation protocol was identified that combines all of the following:

- a graph-attention MARL action explained at decision level;
- paired clean and stale versions of the same operational observation;
- explicit measurement of attention placed on stale telemetry;
- a control for graph nodes that also remove available actions; and
- replication across independently trained checkpoints.

The contribution is therefore an audit protocol and an empirical test, not a new dispatch algorithm. SUMO supplies a controlled simulated state (Lopez et al., 2018). The GAT- MAPPO policy supplies a realistic attention channel. The research contribution is to test whether that channel gives a reproducible freshness-aware and decision-relevant explanation, and to state clearly when the evidence cannot support that claim.

Chapter 3: Methodology

3.1 Study design

The study uses a controlled simulation because the research question requires the same decision to be observed in two ways: once with current telemetry and once with degraded telemetry. A real operational log cannot provide this exact clean twin. SUMO provides the simulated physical state, while a separate observation layer controls what the policy receives.

The final protocol is defined in configs/experiments/dissertation_v4.toml. The full run contains three stages: training, validation-based checkpoint selection, and held-out evaluation. Faithfulness sweeps are run only after the selected checkpoints are frozen.

3.2 SUMO environment and data

The road network is an OpenStreetMap extract of the Central Park area in Yubei, Chongqing. Roads marked as tunnels supply physically meaningful signal-loss triggers. SUMO simulates 20 taxis and 50 passenger requests during each 1,200-second episode. Taxi dispatch is controlled through TraCI, while SUMO continues to update the true position and speed of every vehicle.

Demand variants are separated into training, validation, and test splits. The policy is fitted on training demand. Candidate checkpoints are compared on validation demand. The final reported runs use held-out test demand. This separation prevents test results from influencing model selection.

SUMO is a simulator rather than a neutral source of observed data (Lopez et al., 2018). Its outputs depend on the chosen road network, generated demand, routes, vehicle behaviour parameters, tunnel locations, and random seeds. The experiment is therefore not described as free from bias. Instead, it uses these assumptions consistently in a paired design. At each sampled decision, the clean and degraded observations share the same underlying SUMO state and trained-policy checkpoint; only the observation-layer telemetry differs. This controls the comparison well enough to isolate the effect of the implemented degradation within this scenario, but it does not make the simulated fleet representative of every real city.

3.3 Observation graph and action space

Each idle taxi receives a self-centred graph containing:

Table 3.1: Observation graph node types and roles

Node type	Main features	Role
Self taxi	position, time, speed, AoI, position-valid flag	acting vehicle
Peer taxi	relative position, availability, distance, AoI	nearby fleet context
Passenger request	pickup/drop-off vectors, waiting time	candidate request and action

The graph contains up to five nearby taxis and five nearby requests. Features are normalised and request/taxi masks prevent padded nodes from receiving attention.

The discrete action space contains no-op plus one action for each visible passenger request. This one-to-one relationship is important: deleting a passenger-request node also removes the associated action, whereas deleting a peer-taxi node only hides information about that taxi. Section 3.7 explains how the faithfulness audit controls this asymmetry.

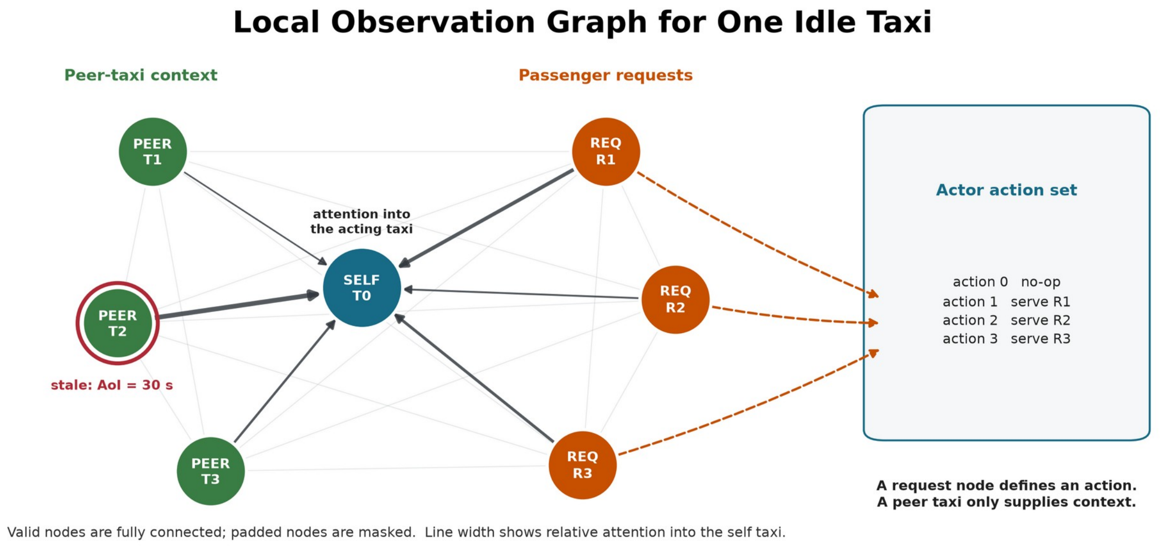

Figure 3.1. Valid self, peer-taxi and request nodes form a fully connected local graph. The red ring marks a stale peer observation, and the illustrative edge widths show attention into the acting taxi. Only request nodes map directly to dispatch actions, which motivates the construct-validity controls.

3.4 Policy models and training

The main policy uses per-node-type feature projections followed by two graph attention layers with four heads. The actor scores no-op and the visible requests. The critic estimates value for MAPPO training with centralised training and decentralised execution. The GAT implementation returns its attention tensors directly so they can be audited without changing the trained network.

The implementation uses scaled dot-product graph attention on the fully connected set of valid local nodes. For node i, node j, layer l, and head h: This differs from the additive attention score in the original GAT paper, so "GAT" in the experiment labels denotes the graph-attention policy family, not an exact reproduction of that layer.

```
q_i = W_Q,h LayerNorm(h_i^l),   k_j = W_K,h LayerNorm(h_j^l)
v_j = W_V,h LayerNorm(h_j^l)
s_ij^(l,h) = (q_i dot k_j) / sqrt(d_head)
alpha_ij^(l,h) = softmax_j(s_ij^(l,h))
z_i = concat_h sum_j alpha_ij^(l,h) v_j
u_i = h_i^l + W_O z_i
h_i^(l+1) = u_i + FFN(LayerNorm(u_i))
```

The mask removes padded nodes before the softmax. A residual connection keeps the previous node embedding. The actor reads the updated self embedding for no-op and each updated request embedding for its matching dispatch action. The default explanation is the self row (i = 0) averaged over both layers and all four heads, then renormalised:

```
alpha_j = normalise((1 / (L H)) sum_l sum_h alpha_0j^(l,h))
```

This aggregation is declared before evaluation. A sensitivity analysis also reports each layer, head-wise values, head maximums, and attention rollout; it does not select an aggregation after seeing which result is favourable. The explanation average is an analysis choice: inside the policy, head outputs are concatenated and projected rather than averaged.

The fixed self row represents the acting taxi's dashboard view and is the explanation channel originally declared for audit. It directly contributes to the no-op embedding. A request-action logit, however, is read from that request's updated embedding, so its corresponding request row is more directly aligned with the selected logit. The revision audit therefore reports a separate request-action sensitivity comparison between the fixed self row and the selected request row. This comparison is not used to redefine the primary metric after seeing the result.

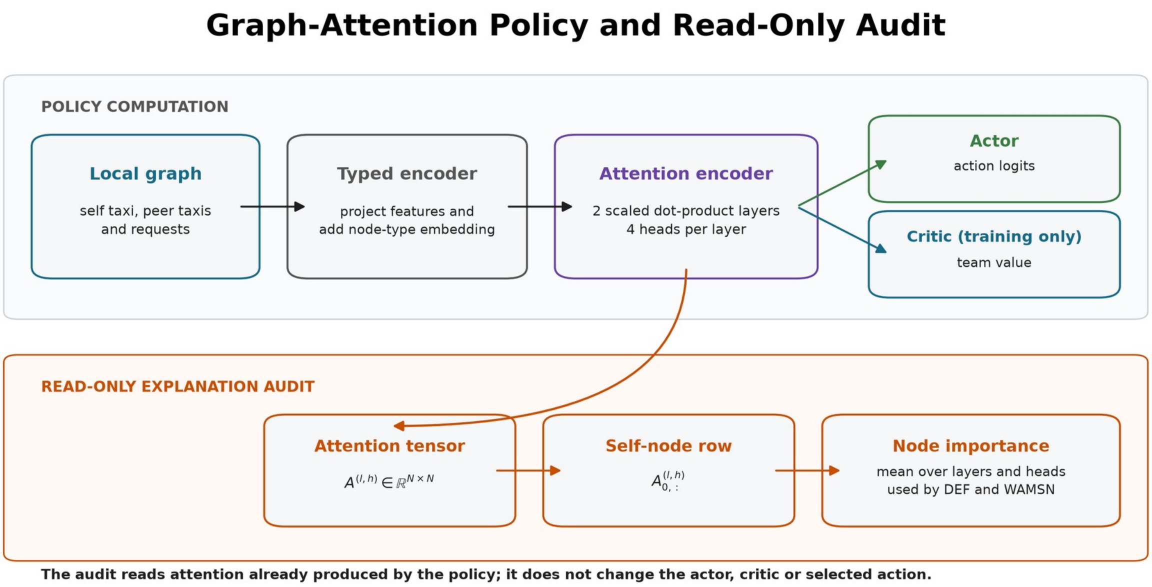

Figure 3.2. The policy and audit share the same two-layer attention encoder. The audit reduces the self-node attention row to node importance without modifying the actor, critic or selected action.

Three policy conditions are included:

Table 3.2: Policy conditions used in the experiment

Model	Policy	Training observations	Purpose
MLP	MLP-MAPPO	clean	performance context without graph attention
GAT	GAT-MAPPO	clean	primary attention model under audit
GAT-Outage	GAT-MAPPO	tunnel-triggered 30-second outages	degradation-aware training condition

Each condition is trained for 150 epochs with seeds 42, 43, and 44.

Candidate checkpoints are saved every ten epochs. Selection uses three stochastic validation episodes with seed 2026 and mean pickups as the primary criterion; mean reward and earlier epoch break ties. The test split is not read during selection. The selected epochs are:

Table 3.3: Validation-selected checkpoint epochs

Model	seed 42	seed 43	seed 44
MLP	50	30	40
GAT	10	70	10
GAT-Outage	10	90	10

The shared optimisation settings are shown below.

Table 3.4: Shared training and optimisation settings

Setting	Value
Training epochs	150
Learning rate	0.0003
Discount factor (`gamma`)	0.99
GAE factor (`lambda`)	0.95
PPO clip ratio	0.20
PPO passes per update	4
Minibatch size	256
Value-loss coefficient	0.50
Entropy coefficient	0.01
Gradient-norm limit	0.50
GAT hidden width / layers / heads	64 / 2 / 4

At simulation step t, the shared team reward is:

```
r_t = 10 N_pickup,t + 0.5 N_dispatch,t - 0.001 mean_wait_t
```

The reward trains dispatch behaviour. It contains no term that rewards a faithful, stable, or human-readable attention map.

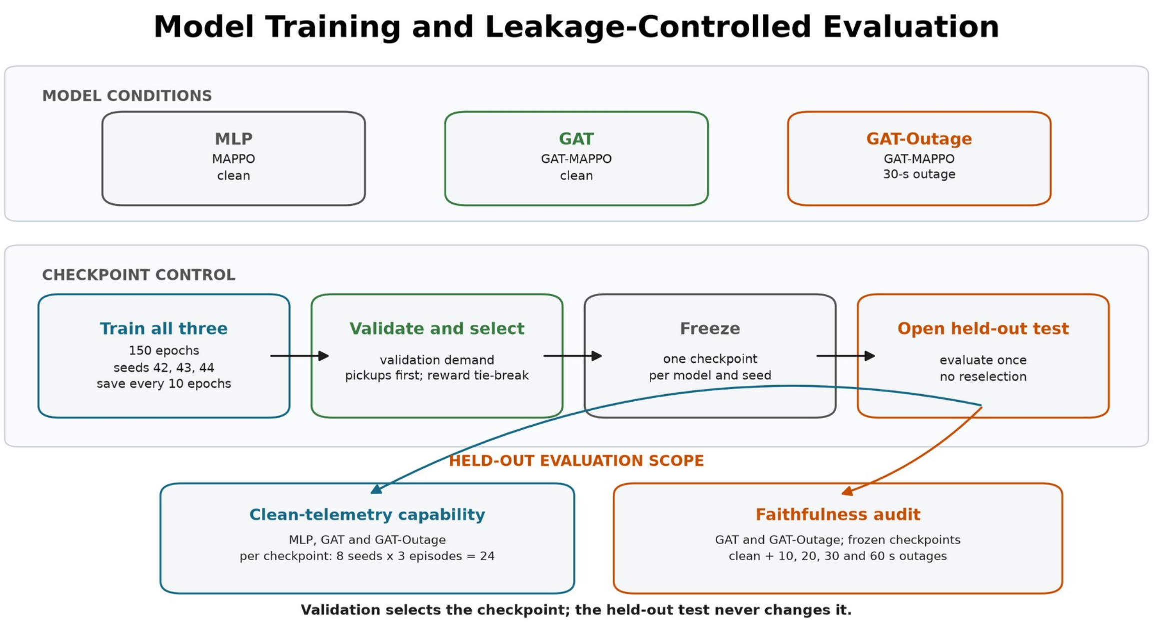

Figure 3.3. Common training and validation-based checkpoint selection for all three model conditions. Each frozen checkpoint receives 24 clean held-out episodes. The faithfulness audit then compares the frozen GAT and GAT-Outage checkpoints across clean and four outage conditions without reselection.

3.5 Telemetry degradation

The trigger and the degradation location are separate parts of the design.

Step 1 - Trigger: when a taxi enters a tunnel edge, the event starts one outage.

Step 2 - Observation-layer outage: for the declared duration, the policy sees the taxi's last valid position and speed. SUMO continues to simulate the true state.

Step 3 - Recovery: after the fixed window expires, a current reading is accepted. A taxi must leave and enter a tunnel again before another tunnel-entry event can trigger.

The fixed outage durations are 10, 20, 30, and 60 seconds. They are easy to monitor and compare, and they create increasing empirical degradation rates. They are not treated as direct causal doses of explanation faithfulness.

AoI is calculated as the current simulation time minus the time of the last valid update and is included in each GAT observation. WAMSN uses normalised AoI as a staleness weight. For this reason, an increase in WAMSN with outage duration partly verifies that the manipulation created longer stale exposure; it is not by itself proof that AoI changed DEF.

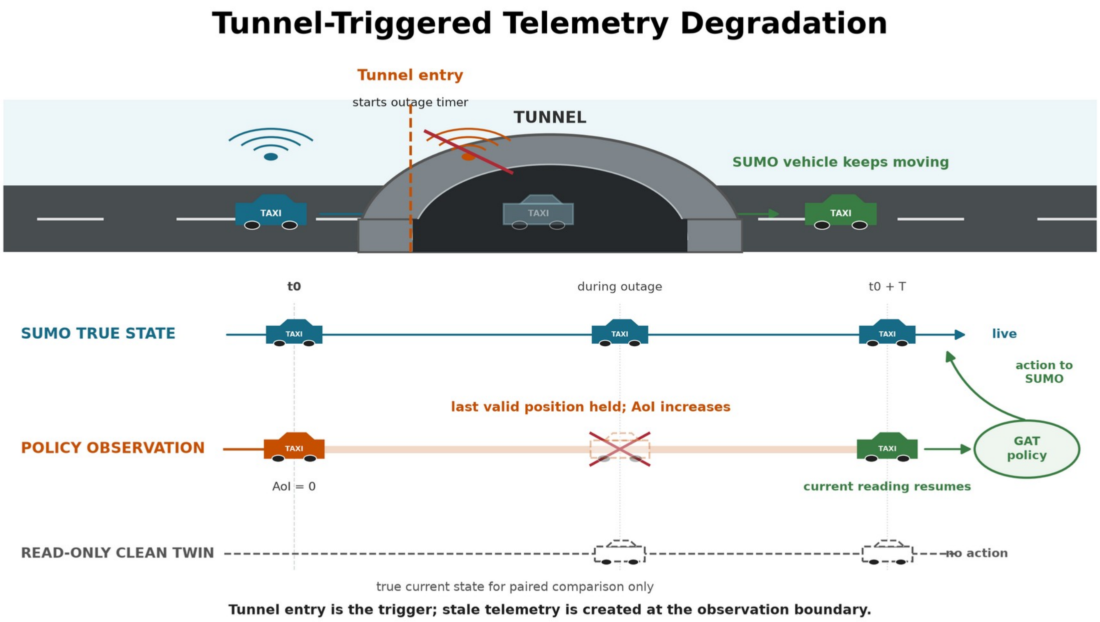

Figure 3.4. Tunnel entry triggers an observation-layer outage. SUMO keeps moving the vehicle while the policy sees its last valid position; the clean twin reads current state but never acts.

3.6 Explanation measures

3.6.1 Decision-level explanation faithfulness

DEF asks whether the nodes ranked highly by an explanation affect the chosen action more than a size- and type-matched random set. For top-k nodes R_k, comprehensiveness measures the output change when those nodes are removed; sufficiency measures the output retained when only those nodes remain. The two gains are compared with five matched random subsets for each k in {1,2,3} and then averaged.

Let m(G,a) be the selected action's logit minus its strongest available alternative. For explanation subset R_k and a matched random subset B_k:

```
Comp(R_k) = m(G,a) - m(G without R_k,a)
Suff(R_k) = m(G,a) - m(G keeping only R_k,a)
g_comp(k) = Comp(R_k) - mean_b Comp(B_k,b)
g_suff(k) = mean_b Suff(B_k,b) - Suff(R_k)
DEF = mean_k 0.5 [g_comp(k) + g_suff(k)]
```

Comprehensiveness is better when removing the explanation weakens the action more than removing random nodes. Sufficiency is better when keeping the explanation loses less evidence than keeping random nodes. The logit margin is clipped to [-10, 10] only when an action becomes unavailable or unopposed; every clamp is logged.

Type-matched random subsets are the **control baseline**; they are not themselves a faithfulness score. Corrected DEF is the measured difference between the attention-ranked subset and this matched random control. Its sign is interpreted as follows:

1. **Positive DEF:** the attention-ranked nodes affect the selected action more than comparable random nodes. This supports decision-level faithfulness.
2. **DEF near zero:** the attention ranking has no measured advantage over the matched random control.
3. **Negative DEF:** the attention-ranked nodes affect the selected action less than the matched random nodes. This does not support the attention ranking as a faithful explanation.

These interpretations apply to both probability DEF and the action-protected logit-margin DEF. The final hypothesis tests use the type-matched baseline described in Section 3.7. DEF provides comparative perturbation evidence; even a positive value does not prove that attention is a complete causal explanation.

3.6.2 Stale-node attention

WAMSN measures attention attached to stale vehicle information:

```
WAMSN = sum(attention_i * normalised_AoI_i) / sum(attention_i)
```

Only self and peer-taxi nodes contribute because request nodes do not carry vehicle telemetry. The analysis reports WAMSN when at least one stale vehicle node is visible, together with stale-node attention share, stale-node mass, and whether a stale node enters the top three.

The paired stale-attention shift compares the degraded observation with its exact clean twin at the same simulation step. Positive values mean the degraded observation assigns more attention mass to nodes marked stale; negative values mean it assigns less.

3.7 Construct-validity audit

The audit supports the primary research problem by checking whether DEF is a valid comparison in this action-candidate graph.

A uniform random occlusion can select passenger requests more often than the attention top-k. Removing such a request changes both the observation and the available action set. A nearby-taxi occlusion changes only information. The result can therefore reflect different node-type composition rather than explanation quality.

The corrected protocol uses three controls:

- **type matching:** random subsets contain the same mixture of taxi and request nodes as the explanation subset;
- **chosen-action protection:** the request associated with the selected action is not deleted in the action-protected variant;
- **clamp logging:** counterfactual action-margin clamps are counted so that action deletion can be detected.

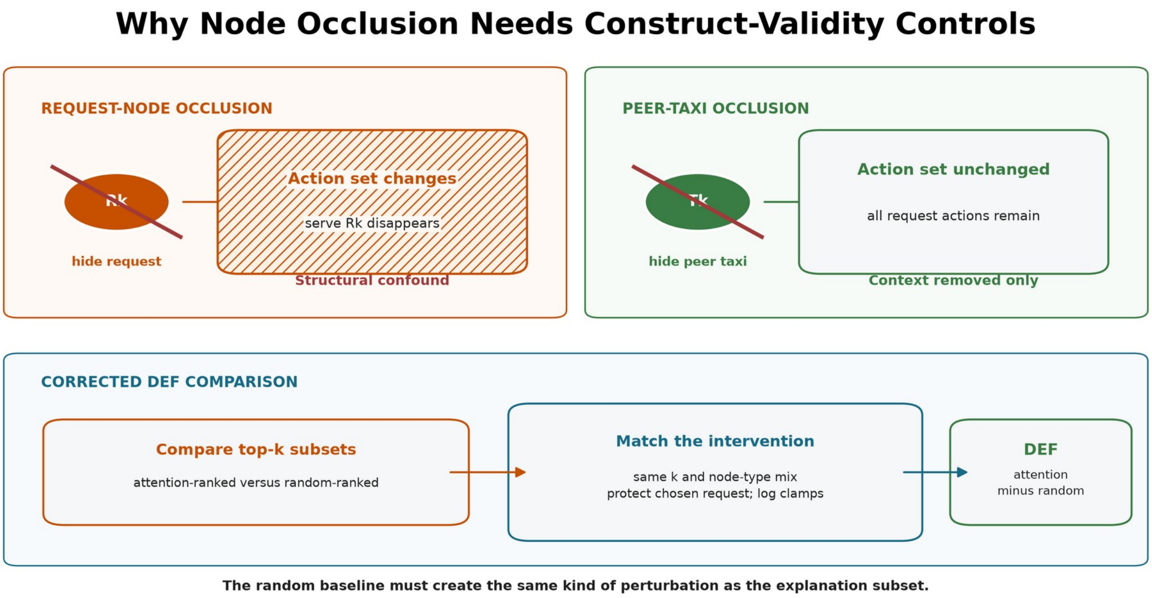

Figure 3.5. Request-node occlusion can remove its action, while peer-taxi occlusion removes context only. Corrected DEF matches subset size and node types, protects the chosen request, and logs margin clamps.

The earlier uniform-baseline results are retained only as an audit trail. They are not used for the final hypothesis verdicts.

Three additional diagnostics test whether a near-zero DEF can be interpreted. First, a leave-one-out (LOO) control masks each valid node separately and ranks nodes by the resulting selected-action margin loss. The selected request is protected. This control is expected to strengthen comprehensiveness because it is built from the same single-node perturbation, but it is not an independent proof that the combined DEF is optimal. Second, Gradient x Input ranks each node by the L2 norm of the selected-action logit gradient multiplied by its raw features. It is a post-hoc comparator that does not use attention weights. Third, the audit reports valid taxi/request counts and the exact expected top-k overlap with a type-matched random subset. Taxi-only Spearman correlation between attention and LOO effects supplies a ranking measure that does not delete request actions and does not depend on random subset overlap.

3.8 Evaluation matrix

MLP, GAT, and GAT-Outage are evaluated under clean telemetry with eight evaluation seeds (42-49) and three episodes per seed. One checkpoint is frozen for each model and training seed, so this gives 24 held-out episodes per checkpoint. The clean capability evaluation therefore contains 216 episode evaluations:

```
3 models x 3 frozen checkpoints per model x 8 evaluation seeds x 3
episodes
```

The complete evaluation structure is summarised below. An episode evaluation is one complete rollout of one frozen checkpoint under one telemetry condition.

Table 3.5: Held-out evaluation structure and episode counts

Evaluation	Models	Checkpoints	Conditions	Episodes per cell	Total episodes
Clean capability context	3	9	1	24	216
Faithfulness severity sweep	2	6	5	24	720
Added ranking controls	2	6	2	24	288

GAT and GAT-Outage receive the full faithfulness sweep:

```
3 training seeds x 5 conditions x 8 evaluation seeds x 3 episodes
```

The five conditions and their roles are:

Table 3.6: Telemetry conditions used in held-out evaluation

Condition	Observation-layer treatment	GAT-Outage relation	Evaluation purpose
Clean	Current vehicle readings	No imposed outage	Capability context and faithfulness baseline
10 s	Freeze last valid reading for 10 seconds	Unseen shorter duration	Mild-degradation transfer test
20 s	Freeze last valid reading for 20 seconds	Unseen shorter duration	Intermediate transfer test
30 s	Freeze last valid reading for 30 seconds	Training-matched duration	In-condition degradation test
60 s	Freeze last valid reading for 60 seconds	Unseen longer duration	Severe stress and transfer test

The relation column refers only to GAT-Outage; the clean-trained GAT has not seen any imposed outage duration during training. The evaluation changes the observation-layer condition but never retrains or reselects a checkpoint. Faithfulness is sampled every eight decisions, with raw per-decision records retained. Each sweep must pass preflight checks for expected cells, provenance, type-matched controls, chosen-action exclusion, and empirical degradation differentiation. All six evaluation sweeps pass.

3.9 Hypotheses and statistics

The confirmatory hypotheses are:

- **H1:** DEF decreases as outage duration increases.
- **H2:** faithfulness declines faster than dispatch performance.
- **H3:** WAMSN increases as outage duration increases.
- **H4:** decisions with greater WAMSN have lower DEF within an episode.
- **H5:** degradation-aware training changes or mitigates these relationships.

H1 and H3 use episode-block permutation trend tests and cluster bootstrap confidence intervals. H2 uses paired cell-level sign flips relative to each evaluation seed's clean condition. H4 calculates Spearman correlation within each episode before a sign-flip test. Holm correction is applied to H1-H4 within each trained policy.

H2 is retained as a pre-declared but exploratory comparison. Its original rate divides by clean DEF, which is close to zero and can make a small absolute change look arbitrarily large. The analysis therefore reports the support verdict and absolute trajectories, but does not use the ratio as the main evidence for faithfulness decoupling.

Thousands of decisions improve measurement precision but do not replace independent model replication. The final synthesis therefore treats each training seed as one trained-policy replicate. With only three seeds per model, the dissertation reports the range and number of supporting seeds and does not claim a cross-seed population p-value.

The added ranking controls use the same held-out demand and stochastic action protocol. They evaluate GAT and GAT-Outage under clean telemetry and a 60-second outage:

```
2 models x 3 training seeds x 2 conditions x 8 evaluation seeds x 3
episodes
```

LOO, Gradient x Input, overlap, and aggregation diagnostics are sampled every eight decisions. Query-row sensitivity is evaluated at every selected-request action because it asks whether the explanation row matches the node that produces that action logit. Decision records are averaged within episodes before bootstrap confidence intervals are calculated. Attention-aggregation sensitivity is reported per training seed and only for decisions where at least one stale vehicle node is visible.

3.10 Reproducibility

The experiment runner records the configuration, commands, seeds, checkpoint hashes, source revision, and preflight reports under the local runs/dissertation_v4/ directory. Compact thesis-ready outputs are committed under results/dissertation_v4/: summary.csv, performance_context.csv, and training_seed_synthesis.csv. The last file makes training-seed consistency explicit. Reproduction commands and expected outputs are documented in docs/REPRODUCE_EXPERIMENTS.md.

3.11 Ethics and data governance

The experiment uses a simulated taxi fleet, generated passenger demand, and an OpenStreetMap-derived road network. It contains no human participants, personal passenger records, live vehicle identifiers, or commercial dispatch data. The study evaluates model behaviour rather than people. Its main ethical risk is miscommunication: presenting an attention map as a trustworthy reason could give an operator false confidence. The reporting therefore separates freshness exposure, task behaviour, and explanation faithfulness, and states negative or mixed findings without turning them into stronger causal claims.

Chapter 4: Results

The primary experiment outputs are stored in results/dissertation_v4/. The additional baseline and faithfulness-control outputs are stored in results/dissertation_revision_v1/. The source audit passed all 100 provenance and protocol checks. Results are reported by training seed because decisions and episodes from one checkpoint do not replace independent model replication.

4.1 Policy capability and training stability

Validation selected GAT epochs 10, 70, and 10 and GAT-Outage epochs 10, 90, and 10 for seeds 42, 43, and 44. A fixed final epoch would therefore compare different stages of learning.

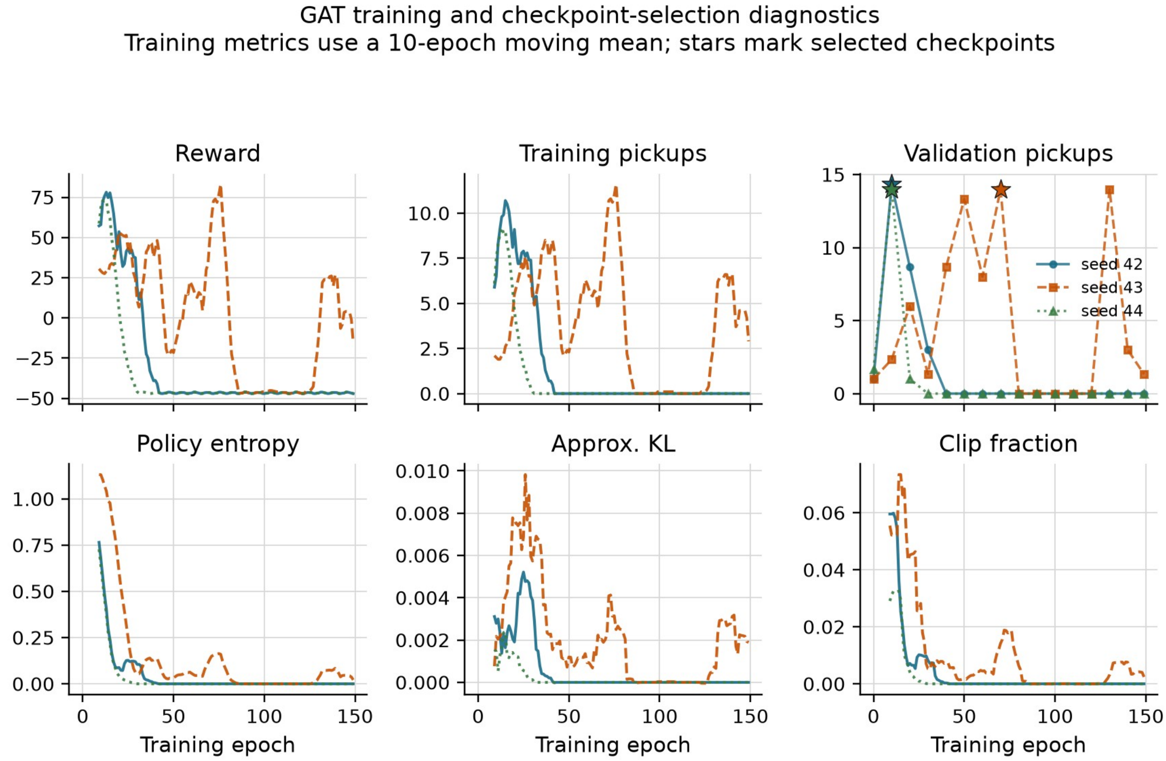

Figure 4.1. GAT training metrics use a ten-epoch moving mean. Stars mark the validation-selected checkpoints. Entropy falls early and the final policies for seeds 42 and 44 collapse to zero pickups.

Unsmoothed GAT entropy first falls below 0.05 at epochs 15, 21, and 12. At the selected checkpoints it is 0.140, 0.246, and 0.136, but at epoch 150 it is 0.000, 0.005, and 0.000, with zero pickups for all three runs. Validation selection avoids presenting a failed final checkpoint as the audited policy. It does not show that optimisation is stable.

Table 4.1 places the selected policies above two lower bounds evaluated on the same held-out demand. Performance is reported only to establish that the audit does not concern a random-level policy.

Table 4.1: Policy capability on held-out demand

Policy	Mean pickups	Uncertainty or replicate range
Legal random	1.50	95% episode-cluster CI [1.08, 1.96]
Greedy nearest request	2.00	95% demand-variant CI [1.00, 3.00]
MLP	13.25	training-seed range 11.50-14.67

Policy	Mean pickups	Uncertainty or replicate range
GAT	13.29	training-seed range 11.46-14.67
GAT-Outage	9.56	training-seed range 4.54-12.88

The legal-random baseline has 24 independent episode clusters. Greedy nearest is deterministic, so its effective sample is the three demand variants rather than 24 repeated seed-episode records. GAT's seed means are 14.67, 13.75, and 11.46. It clearly exceeds both lower bounds, but its similarity to MLP does not show that graph attention caused the task capability. GAT-Outage seed 44 remains a weak replicate at 4.54 pickups and is retained rather than excluded.

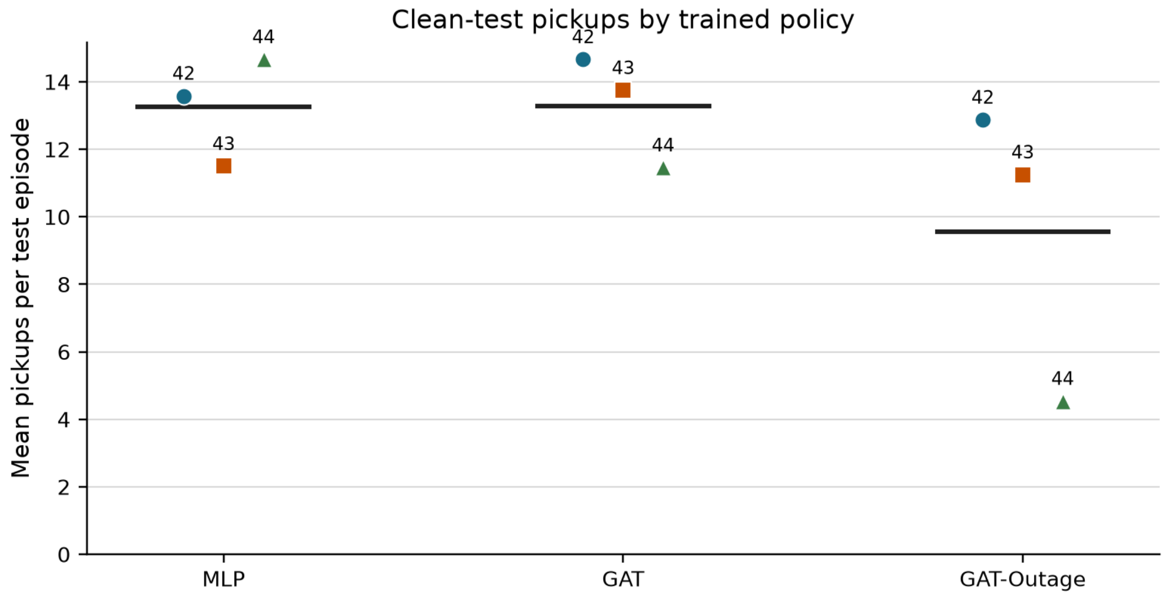

Figure 4.2. Each point is one selected policy evaluated over 24 held-out episodes. Horizontal lines show the cross-seed mean.

4.2 Construct-validity audit

Removing a request node can remove an available action, while removing a peer taxi only hides information. A uniform random occlusion therefore compares unlike interventions.

Table 4.2: Construct-validity audit of random controls

Control baseline used to calculate DEF	Mean margin-DEF	95% CI	Interpretation
Uniform random	-0.5540	[-0.5808, -0.5288]	dominated by node-type and action-deletion imbalance
Type-matched random	-0.0093	[-0.0159, -0.0013]	small negative result after matching node types

The type-matched interval excludes zero, but its magnitude is about two orders smaller than the uniform-control artefact. This earlier 1,199-decision audit supports the measurement design; it is not a separate answer to the primary research question. The final protocol uses type matching, protects the chosen request in the margin variant, and logs action-margin clamps.

4.3 Evaluator sensitivity and ranking controls

The revision evaluates raw attention, Gradient x Input, and an LOO perturbation ranking under clean and 60-second conditions. Each model, training seed, and condition contains eight evaluation seeds and three episodes. Table 4.3 shows clean results. For each checkpoint, the 60-second DEF values differ by less than 0.001 from clean; taxi-only rho changes by at most 0.012.

Table 4.3: Clean-telemetry faithfulness controls by training seed

Model	Seed	Raw DEF	GxI DEF	LOO DEF	Taxi-only rho
GAT	42	+0.0001	-0.0050	+0.0007	+0.022
GAT	43	-0.0084	-0.0082	+0.0114	+0.205
GAT	44	-0.0015	-0.0023	+0.0024	-0.411
GAT-Outage	42	+0.0003	-0.0054	+0.0011	+0.319
GAT-Outage	43	+0.0187	-0.0405	+0.0663	-0.710
GAT-Outage	44	-0.0008	-0.0013	+0.0019	-0.261

LOO is positive and larger than raw attention for all six checkpoints. This shows that DEF responds to a ranking built from single-node margin loss. It is still only a perturbation control: its gain is mainly expected in comprehensiveness, and it does not prove that the combined DEF has an ideal upper bound. Gradient x Input is negative for every checkpoint and is not a better explanation under this evaluator.

Raw attention does not have one clean-telemetry result. It ranges from -0.0084 to +0.0187, with mixed signs. Taxi-only rank correlation with LOO ranges from -0.710 to +0.319. The important result is therefore not that attention equals random in every model. It is that its measured decision relevance does not reproduce across independently trained checkpoints.

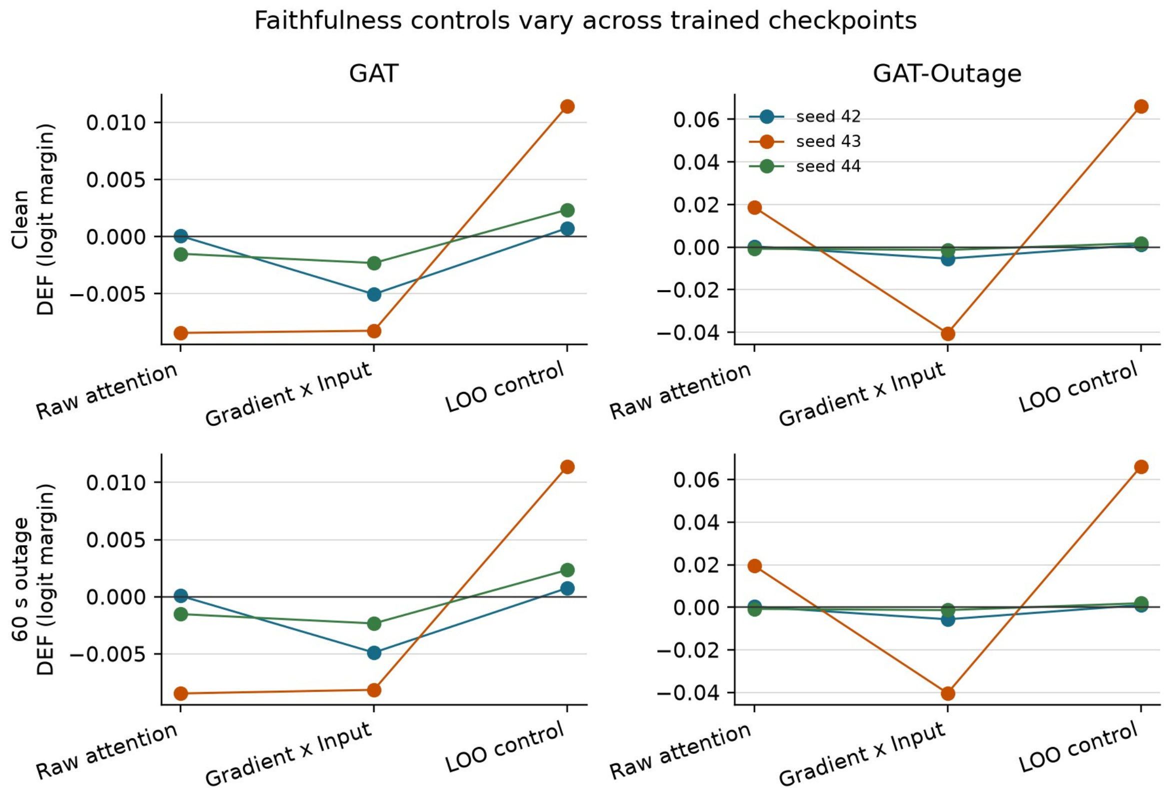

Figure 4.3. Episode means are first calculated within each checkpoint. Lines identify training seeds; they are not population trends. LOO is positive for every checkpoint, while raw attention and Gradient x Input vary.

4.4 Small-graph resolution

The scored graphs usually contain the maximum number of visible nodes.

Table 4.4: Valid node-count distribution in scored graphs

Model	Node count	Mean	Median	5th percentile	95th percentile
GAT	peer taxis	5.00	5	5	5
GAT	requests	4.61	5	1	5
GAT	non-self total	9.61	10	6	10
GAT-Outage	peer taxis	4.99	5	5	5
GAT-Outage	requests	4.64	5	1	5
GAT-Outage	non-self total	9.63	10	6	10

Even so, type matching creates substantial top-k overlap. The expected random intersection is about 0.21 for k=1, 0.41 for k=2, and 0.61 for k=3. Attention-LOO agreement at k=3 is 0.48 for GAT and 0.52 for GAT-Outage, below the expected random overlap. Restricting analysis to at least six non-self nodes does not change the sample because every scored decision already meets that threshold.

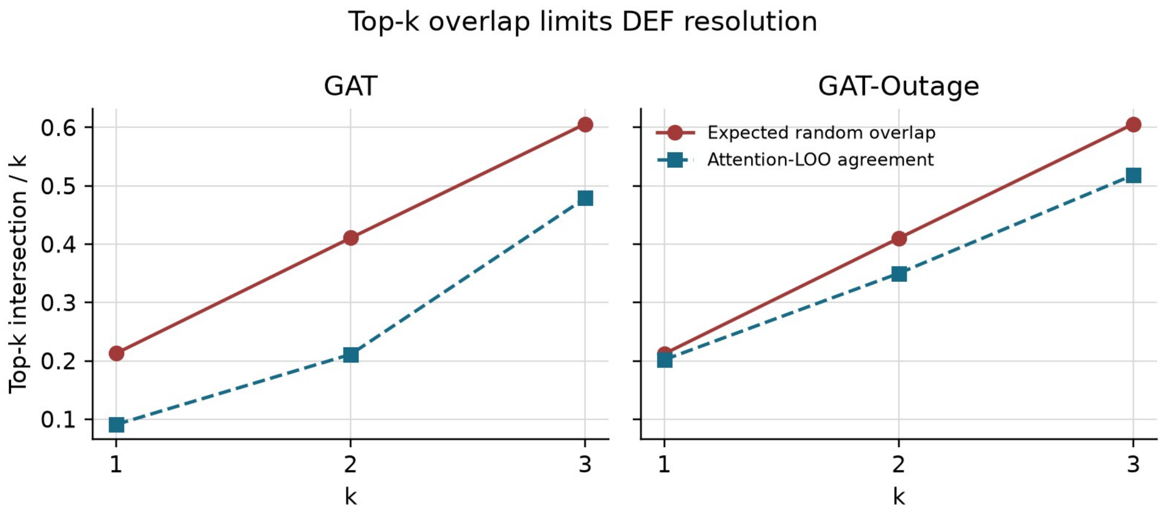

Figure 4.4. Large expected overlap between the explanation and its matched random set compresses DEF, especially at k=3.

4.5 Telemetry manipulation and stale exposure

Tunnel-triggered observation outages create increasing stale exposure. At 60 seconds, the degraded-decision rate is about 0.9-1.2% for GAT and 0.7-1.3% for GAT-Outage. Exposure is sparse because it requires a taxi to enter a mapped tunnel.

Across the three checkpoints, stale-exposed decision counts for GAT are 86, 106, 182, and 266 at 10, 20, 30, and 60 seconds. GAT-Outage counts are 87, 101, 181, and 259. Conditional WAMSN rises from about 0.023-0.031 at 10 seconds to 0.071-0.090 at 60 seconds. H3 is supported for all six policies. Since WAMSN contains normalised AoI, this is a manipulation and exposure result, not proof that AoI causes a change in faithfulness.

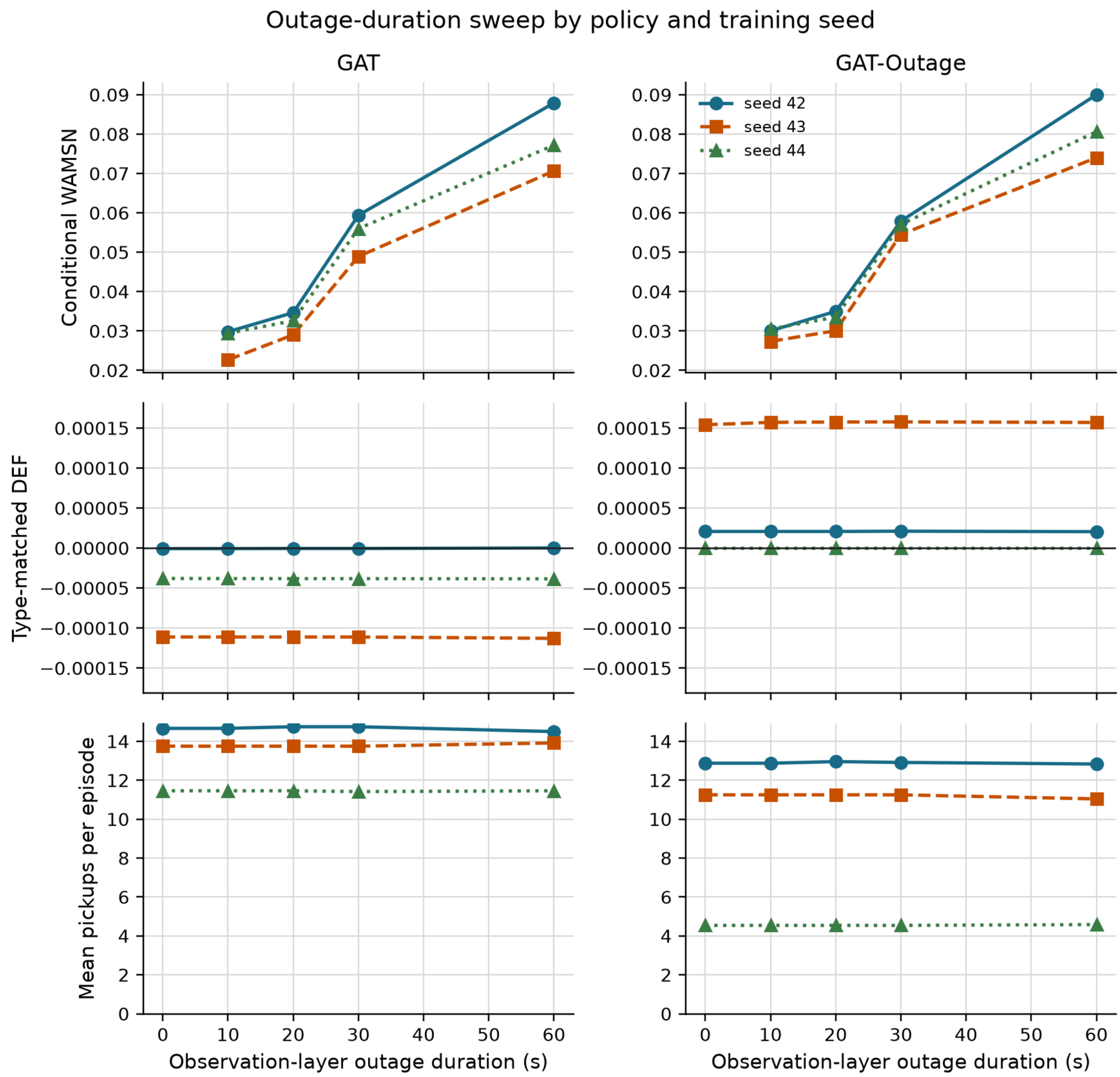

Figure 4.5. Conditional WAMSN rises with outage duration, while DEF and pickups show little movement. Pickups are supporting context, not the research outcome.

4.6 Attention reallocation and aggregation sensitivity

The declared two-layer, four-head mean gives opposite paired stale-attention shifts across training seeds.

Table 4.5: Paired stale-attention shift estimates and confidence intervals

Model	Training seed	Mean shift	95% CI
GAT	42	+0.002706	[0.002395, 0.003036]
GAT	43	+0.000618	[0.000424, 0.000780]
GAT	44	-0.001278	[-0.001436, -0.001112]
GAT-Outage	42	+0.002374	[0.002151, 0.002609]
GAT-Outage	43	+0.000783	[0.000710, 0.000862]
GAT-Outage	44	-0.001231	[-0.001410, -0.001072]

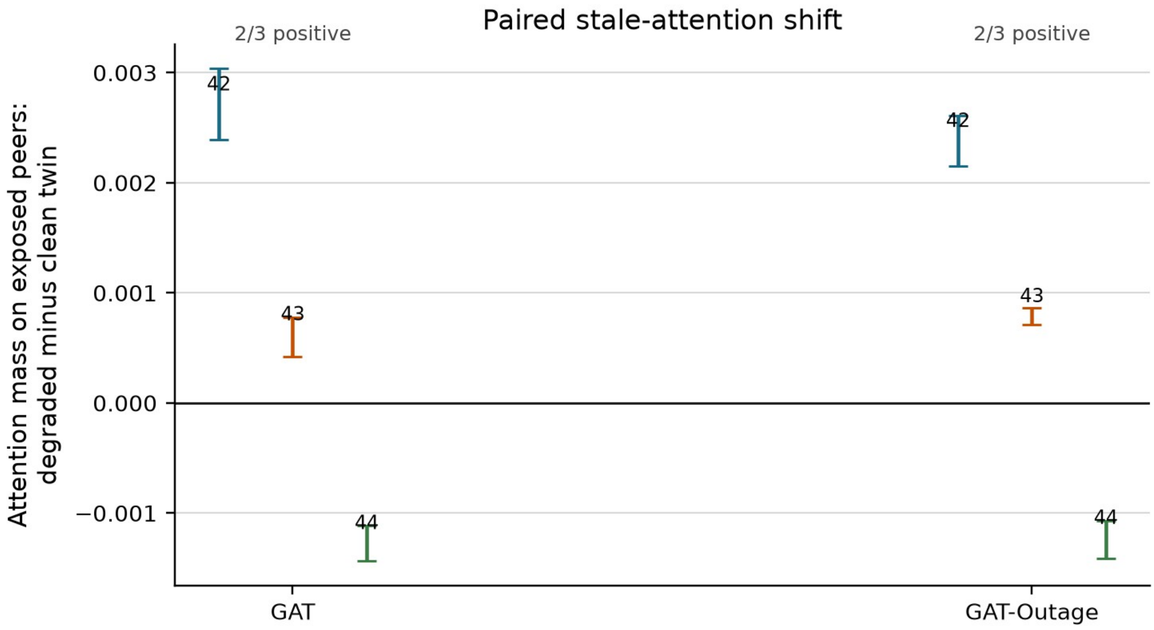

Figure 4.6. The primary aggregation shifts toward stale nodes for seeds 42 and 43 and away from them for seed 44 in both training regimes.

Alternative aggregation rules make the dependence stronger. Across the 13 layer, head, maximum, and rollout choices, every checkpoint has at least one positive and one negative result. For example, GAT seed 43 ranges from -0.0060 to +0.0067, while GAT-Outage seed 43 ranges from -0.0020 to +0.0033. GAT seed 44 and GAT-Outage seed 44 are mainly negative, but individual heads remain positive.

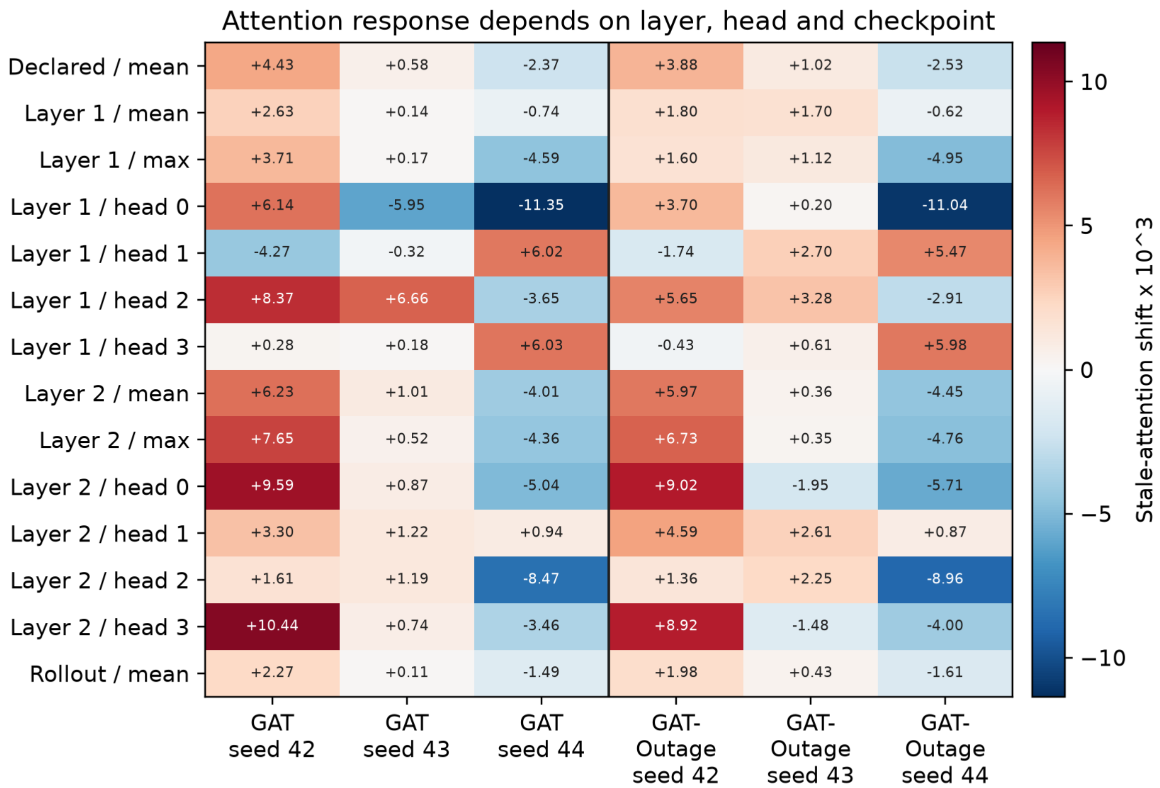

Figure 4.7. Values are degraded-minus-clean stale-attention mass multiplied by 1,000. The sign varies by checkpoint and, in some models, by layer or head.

The query row also matters. For request actions, changing from the declared self row to the selected request row changes DEF by -0.0005, +0.0069, and -0.0002 for GAT seeds 42-44 under clean telemetry. GAT-Outage changes are -0.0004, -0.0543, and -0.0009. The 60-second values are almost identical. The request row improves one GAT checkpoint but worsens the others, especially GAT-Outage seed 43. It does not provide a general correction for the primary explanation channel.

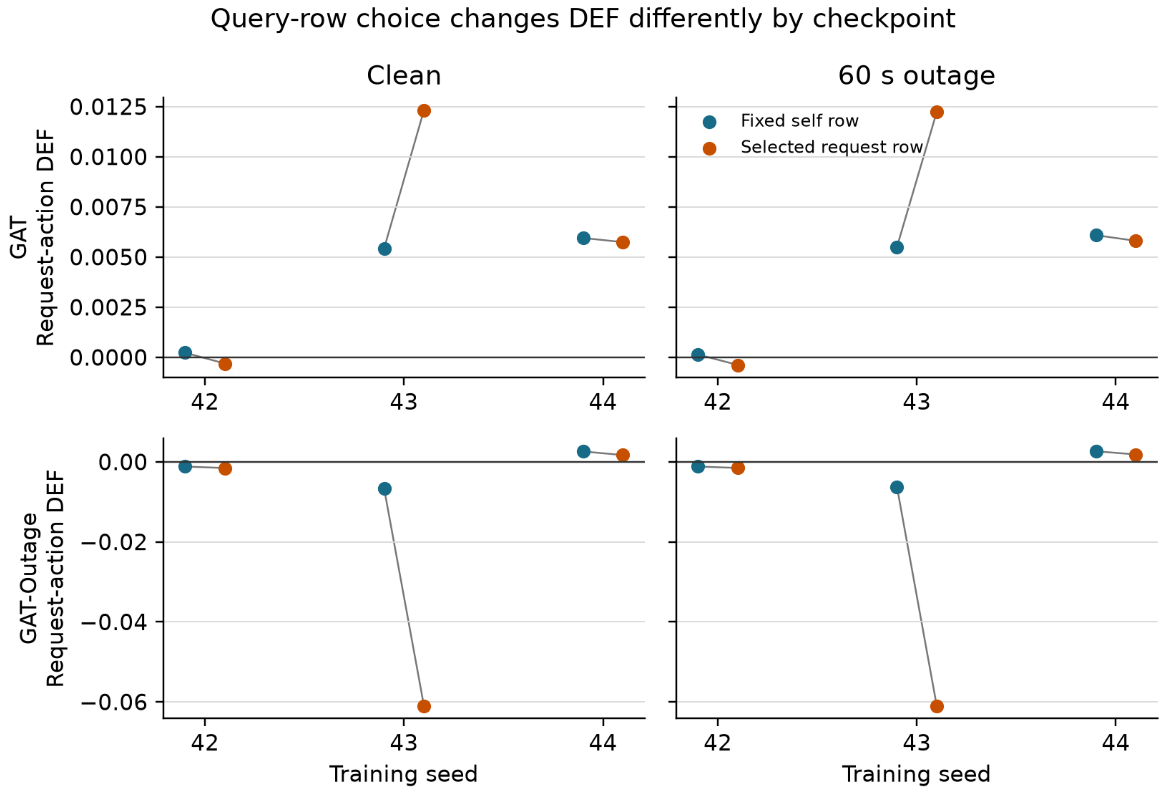

Figure 4.8. Each grey line joins the self-row and selected-request-row DEF for one checkpoint. The effect of changing query row is checkpoint-dependent.

4.7 Faithfulness hypotheses

H1 is unsupported for every GAT and GAT-Outage checkpoint. DEF-duration correlations remain close to zero, with Holm-adjusted p-values above 0.67. H2 is also unsupported: DEF does not decline faster than pickups. Because clean DEF can be near zero, the original ratio is unstable and is kept only as an exploratory, pre-declared check.

H4 is mixed. GAT within-episode WAMSN-DEF correlations are positive (0.036-0.069), contrary to the prediction. GAT-Outage seeds 42 and 43 are negative (-0.072 and -0.061, adjusted p=0.0004), but seed 44 is approximately zero.

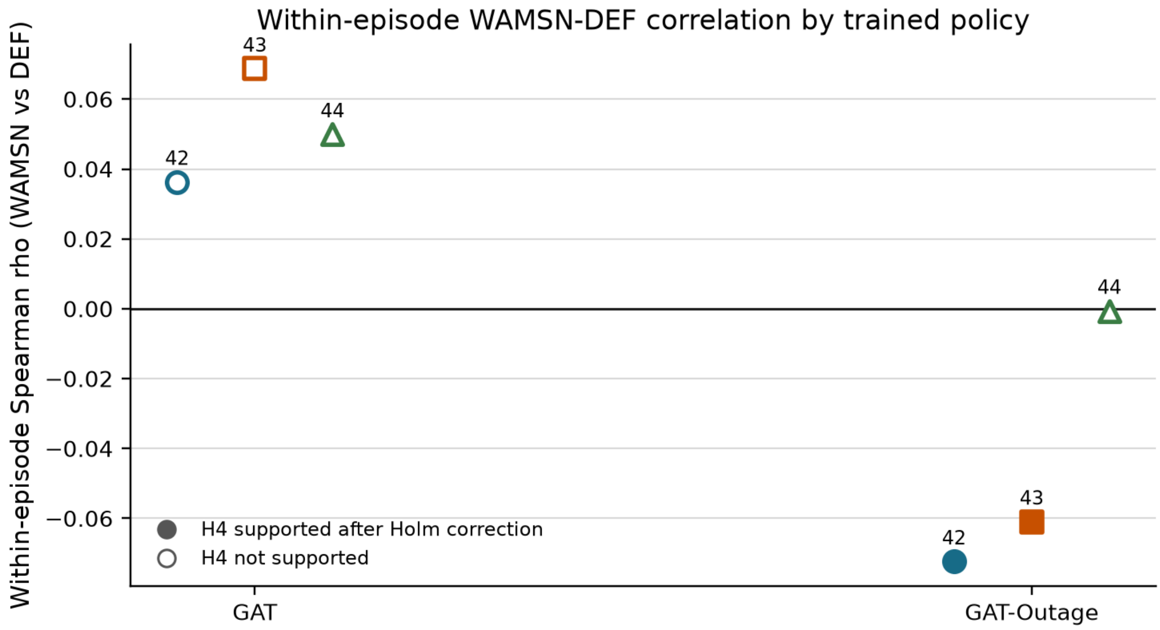

Figure 4.9. Filled markers pass the within-checkpoint Holm-corrected H4 test. These tests do not establish consistency across trained policies.

Table 4.6: Hypothesis outcomes across training seeds

Hypothesis	GAT seeds supporting	GAT-Outage seeds supporting	Verdict
H1: outage duration increases, DEF decreases	0/3	0/3	not supported
H2: faithfulness declines faster than performance	0/3	0/3	not supported; exploratory
H3: outage duration increases, WAMSN increases	3/3	3/3	consistently supported
H4: higher WAMSN is associated with lower DEF	0/3	2/3	mixed
H5: degradation-aware training mitigates the effect	n/a	0/3	not supported

4.8 Result summary

The experiment establishes three points. First, the evaluator can detect a positive LOO perturbation ranking, but top-k overlap limits its resolution. Second, longer outages reliably increase stale-data exposure, but not DEF loss. Third, the direction and measured faithfulness of attention depend on the trained checkpoint, aggregation rule, and query row. GAT-Outage does not remove these dependencies. Raw attention is therefore not validated as a stable assurance of either freshness or decision relevance.

Chapter 5: Discussion

5.1 Answer to the central problem

The central problem is whether a complete graph-attention map can be trusted when some of its vehicle data are stale. The answer from this experiment is: **not without separate validation**. Raw attention is not shown to provide a stable assurance of freshness or decision relevance.

This answer does not depend on one expected hypothesis being confirmed. Under clean telemetry, raw-attention DEF has mixed signs and ranges from -0.0084 to +0.0187 across checkpoints. Under a 60-second outage, each checkpoint remains close to its own clean value. Longer outages increase stale-data exposure, but do not produce a consistent loss of DEF. Attention reallocation is positive for two seeds and negative for one under both training regimes.

The added controls make the interpretation stronger and narrower. LOO produces a positive DEF for every checkpoint, so the evaluator has some sensitivity to a perturbation-based ranking. However, expected top-k overlap reaches about 61% at k=3, which limits resolution. Taxi-only attention-LOO correlation is also inconsistent. The result is not "the evaluator is perfect and attention fails." It is "within a measured but limited audit, attention gives no reproducible explanation guarantee."

Cross-seed evidence matrix

Question	Measure	Cross-seed result	Verdict
RQ1	Clean DEF	Near type-matched random baseline	**Not validated**
RQ2	Paired attention shift	Positive in 4/6; negative in 2/6	**Seed-dependent**
RQ3	DEF and pickups	H1 and H2 supported in 0/6 policies	**No decline found**
RQ4	Degradation training	No consistent mitigation	**Not supported**

Conclusion: raw attention provides no reproducible faithfulness guarantee across trained policies.

Figure 5.1. The evidence separates policy capability, evaluator validity, stale exposure, attention response, and d ecision relevance. No single arrow is treated as a causal chain.

5.2 What the stale-exposure result means

WAMSN rises with outage duration for every GAT and GAT-Outage checkpoint. It is useful as an operational exposure indicator: it shows when an attention map contains more weight attached to old vehicle readings.

WAMSN is not evidence that AoI causes lower faithfulness. The metric itself weights attention by normalised AoI, so part of the increase follows directly from the manipulation. The paired stale-attention shift asks whether the model reallocates attention to stale nodes; that result changes sign across seeds. DEF asks whether ranked nodes are decision-relevant; it does not decline with duration. These three measures answer different questions and should remain separate.

5.3 Dependence on checkpoint and analysis choice

The primary aggregation gives positive stale-attention shifts for seeds 42 and 43 and negative shifts for seed 44. The values are small: approximately -0.0013 to +0.0027 of total attention mass. An operator may not perceive changes below three tenths of one percentage point. One reasonable interpretation is that all three are practically close to zero, even though the paired estimates have narrow intervals.

The aggregation audit reveals a second problem. Individual heads can reverse the sign within the same checkpoint. For GAT seed 43, alternative results span about -0.0060 to +0.0067. An analyst could therefore obtain a different verbal conclusion by selecting another head or layer after seeing the data. The declared mean remains the primary result, but the sensitivity analysis shows that it is not a unique explanation produced by the network.

Query-row choice is similarly checkpoint-dependent. The selected request row improves request-action DEF for GAT seed 43, changes little for four checkpoints, and sharply reduces DEF for GAT-Outage seed 43. This rules out a simple explanation that near-zero or negative DEF occurred only because the audit used the self row. More broadly, an operator-facing attention map needs a declared and tested rule for choosing its query, layer, and heads.

5.4 Role of the construct-validity audit

The construct-validity audit supports the primary experiment. Request nodes are both information and actions; peer taxis are context only. Uniform random deletion therefore creates a large action-removal artefact. Type matching and chosen-request protection make the comparison fairer.

The LOO control checks a different question: can DEF respond when nodes are ranked by their own single-node margin loss? Its consistently positive result shows limited sensitivity, mainly for comprehensiveness. It is not an oracle for the combined comprehensiveness-sufficiency score. Gradient x Input is negative in every checkpoint, showing that a familiar post-hoc method is not automatically a stronger explanation in this task.

Together, these checks change the conclusion from a simple “attention is near random” statement to a more defensible one. Raw attention varies in sign and size, remains below the LOO control, correlates inconsistently with taxi-node effects, and is sensitive to analysis choices. None of these observations provides a stable default assurance.

5.5 Degradation-aware training

Training with 30-second outages does not solve the explanation problem. GAT-Outage retains the same seed-dependent primary attention shift as GAT. Its clean raw DEF ranges from -0.0008 to +0.0187, and its taxi-only correlation ranges from -0.710 to +0.319. Query-row sensitivity is especially large for seed 43.

This does not show that all robustness training is ineffective. The reward contains no term for explanation stability or faithfulness, so there is no reason to expect task-reward optimisation alone to align attention with a human explanation. A future intervention would need an explicit explanation objective and evaluation on independently trained models.

5.6 Practical implications

An operator interface should not label raw graph-attention weights as reasons without additional evidence. A deployment should:

- display telemetry freshness separately from attention strength;
- monitor stale-node exposure without treating it as a faithfulness score;
- declare the query row and layer/head aggregation used in the interface;
- run action-aware faithfulness tests for every released checkpoint;
- repeat the audit after retraining, because the qualitative result can change;
- avoid using stable dispatch output as evidence that explanations remain trustworthy.

5.7 Limitations

The study uses one simulated district, 20 taxis, 50 requests, and sparse tunnel exposure. It evaluates three independent training seeds per model. Three seeds are sufficient to expose heterogeneity, but not to estimate a population distribution of training outcomes. GAT-Outage seed 44 is weak, and GAT training shows early entropy collapse. The selected GAT policies outperform random and greedy lower bounds, but the study does not isolate the contribution of graph edges to that capability.

The local graph is small. Type matching makes random top-k sets overlap heavily with the explanation, which compresses DEF. LOO is based on the same single-node perturbation as comprehensiveness and is not an independent ground truth explanation. Other perturbation choices or post-hoc explainers may give different results.

The degradation layer freezes position and speed for fixed windows. Real telemetry may include delayed packets, partial failures, map-matching errors, and asynchronous recovery. The study audits attention as an explanation; it does not claim that attention is useless inside the policy computation.

The paired clean/degraded comparison gives the study internal validity for its implemented observation-layer intervention: both observations are derived from the same simulated traffic state. It does not remove simulation or scenario- design bias. Road-network choice, generated demand, vehicle behaviour, and tunnel placement may affect which taxis become stale and how those taxis are positioned in the local graph. In particular, a fixed tunnel can create an exposure-selection effect if its traffic conditions differ from the rest of the network. The conclusions are therefore limited to the tested SUMO setting and should not be read as population estimates for real fleets or cities.

5.8 Future work

Future experiments should add training seeds and use multiple road networks, demand levels, tunnel placements, and public mobility traces. A randomised telemetry-loss control would help distinguish a general stale-node effect from one produced by tunnel location. Experiments should also include disconnected- edge and message-passing ablations to establish how much the policy uses its graph channel. Delayed, intermittent, and biased telemetry should be compared with fixed freezes.

The explanation audit should also compare LOO with graph-specific post-hoc methods and evaluate objectives that directly reward explanation consistency. Any proposed improvement should be tested on held-out checkpoints, not only on more episodes from one trained model. A human-factors study could then test whether explicit freshness warnings lead operators to interpret the map more appropriately.

Chapter 6: Conclusion

This dissertation examined a practical assurance problem: a graph-attention map can remain visible and appear complete after some vehicle telemetry has become stale. Attention strength alone does not tell an operator whether its source is current or whether the highlighted node affected the action.

The central answer is clear: **this experiment does not validate raw graph-attention weights as dependable explanations under clean or degraded telemetry**. This is not a claim that every attention map is wrong. It means that attention cannot be trusted by default.

The research questions lead to that answer:

- **RQ1: Is attention faithful under clean telemetry?** No stable result is found. Raw-attention DEF ranges from -0.0084 to +0.0187 across checkpoints. LOO gives a larger positive result for all six checkpoints, while taxi-only attention-LOO rank correlation ranges from -0.710 to +0.319.
- **RQ2: Does attention move toward stale vehicle nodes?** Not consistently. The declared aggregation moves toward stale nodes for seeds 42 and 43 and away from them for seed 44 under both training regimes. Individual heads can reverse the direction within one checkpoint.
- **RQ3: Does the explanation remain dependable as outage duration increases?** No dependable relationship is found. Stale exposure increases, but DEF does not consistently decline. The 60-second ranking-control results remain close to their clean values.
- **RQ4: Does degradation-aware training improve explanation reliability?** No consistent mitigation is observed. GAT-Outage retains checkpoint, aggregation, and query-row dependence and includes one weak policy replicate.

The construct-validity audit is essential to this interpretation. Deleting a request can also delete an action, while deleting a peer taxi only hides context. Type-matched and action-protected controls remove much of this artefact. The positive LOO result shows that DEF has some perturbation sensitivity, but the high expected top-k overlap limits its resolution. The conclusion is therefore based on several checks rather than one near-zero number.

In this thesis, **faithfulness decoupling** is the assurance gap between a visible explanation and valid supporting evidence. An attention map can outlive the freshness of its inputs without giving a stable indication of decision relevance. The gap does not require DEF to decline monotonically with outage duration.

The practical resolution is to separate freshness from explanation. Stable dispatch output is not evidence that an explanation remains trustworthy. Attention weights should be treated as model internals until each released checkpoint passes freshness-aware, action-aware, and aggregation-sensitive faithfulness tests, and the result reproduces across independent training runs.

List of Publications

No publication is claimed in this dissertation draft.

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Appendices

Appendix A: Experiment Reproduction

The complete reproduction procedure is provided in docs/REPRODUCE_EXPERIMENTS.md. Compact citable outputs are stored under results/dissertation_v4/.

Appendix B: Supporting Artefacts

The repository contains the experiment configuration, source code, validation selections, preflight reports, per-seed analyses, summary tables and figure sources used in this dissertation. Raw cells and checkpoints remain outside the document because of their size.